# Christos Hadjinikolis

Engineering Manager | Reliable AI Systems & Technical Leadership
Ph.D. Computer Science | UCL Associate Researcher | AI Standards
✉ christos.hadjinikolis@gmail.com

## about me

I am a **hands-on Engineering Manager and ML systems lead**, combining production AI engineering with doctoral research and AI standards work. I lead teams turning research into dependable services, with model/data versioning, evaluation, replay and monitoring built into delivery. Alongside managing people, I continue substantial individual-contributor work on production code and ML/data pipelines. My UCL research and teaching and AI committee work focus on auditability, explainability and responsible adoption.

### profiles
website
LinkedIn
GitHub
Google Scholar

### leadership
6 direct reports
10-person team
hands-on delivery
hiring/coaching

### systems
real-time data
Kafka/Flink
model/API serving
observability

### ML assurance
model/data versioning
evaluation/replay
data contracts
accuracy monitoring

### maturity
ADRs/forums
runbooks/on-call
local E2E tests
rollback/fallback

### domain
AI standardisation
auditability
symbolic reasoning
explainability

### tooling
**PyPI**: dynamicio
skeleton-replay
JetBrains plugin

## work-experience

**Vortexa Ltd, London, UK** 12/2020–present — Engineering Manager | ML Systems Lead

I lead technical direction and delivery for Vortexa's live post-ingestion ML/data estate, turning **6M** vessel-position records/hour into production commodity intelligence for **13.5K** monitored vessels. Manage **6** direct reports and lead a **10-person** cross-functional team accountable for model quality, data-product interfaces, client and stakeholder clarity, reliability, and delivery maturity. Joined as Senior ML Engineer in December 2020; Engineering Manager / ML Systems Lead since February 2022.

- **People & Delivery:** Manage **6** direct reports across Product, SME analysis, Data Science, and Data Engineering; own performance reviews, career development and delivery. Hiring manager for **6** roles; shaped onboarding and hiring practices as Data Production grew from **4 to 30+**. Retained the full team and supported every member's promotion or progression (July 2026 snapshot).
- **Technical Direction:** Shape architecture across model services, data pipelines and evaluation/replay interfaces. Resolve model-quality questions with Product, researchers and domain experts, translating competing priorities into measurable objectives and delivery plans.
- **Production ML Systems:** Led research-to-production delivery for destination and arrival-time sequence/transformer models in **PyTorch**; established MLflow, model/data versioning, evaluation, replay, monitoring and model-serving workflows.
- **Hands-on Implementation:** Led the **Kafka Streams-to-Flink** migration: researched trade-offs, wrote the proposal and personally built the first replacement application into production. Improved operational visibility and compute scaling independent of Kafka partitioning.
- **Operational Assurance:** Established live accuracy monitoring, signal-quality controls, runbooks, rollback/fallback paths and shared on-call practices, so teams could investigate prediction failures and operate evolving models reliably.
- **Engineering Maturity:** Standardised I/O seams, schema/data validations, repo archetypes, ADRs, dev containers, local E2E tests, and DORA/Jira delivery-health signals; cut a **3h** DAG feedback loop to under **5** min.
- **Shared Engineering Practice:** Started a weekly architecture forum to share decisions and reduce knowledge silos. Ran stakeholder workshops and design reviews to make trade-offs explicit and align engineering work with product and client needs.

**DataReply, London, UK** 04/2016–12/2020 — Senior Consultant | Data Scientist / ML Engineer

Joined Data Reply's London spin-off as its first consultant, operating inside and contributing to a practice that grew from **three practitioners and two managers to 30+ consultants**. Progressed to Senior Consultant; shaped opportunities, agreed deliverables/timelines, managed expectations, placed consultants, and helped build the consulting culture around clear technical delivery across UBS, Vodafone, CNHi, and other clients.

- **Vodafone Group:** Led multi-workstream GCP data-science platform work based on Kubeflow; drove requirements, migration planning, CI/CD, workflow automation, ML lifecycle support, and feature-engineering migration.
- **CNHi:** Lead Data Scientist/Scrum Master on predictive maintenance (Azure/Databricks); translated stakeholder needs into DS/DE backlog and guided a **PySpark** team over live telemetry.
- **UBS:** Embedded consultant who grew from Data Scientist into ML Engineer through XP/pairing while delivering graph/process-mining analytics, observability, and real-time **Kafka**/Elasticsearch pipelines.

**KCL, UCL, David Game College, UK** 2010–2016 — Teaching and Technical Training

Delivered courses and tutorials in algorithm design, data structures, databases/SQL, and Java/Python during and after doctoral research; developed presentation discipline, learner support, and mentoring habits.

# education

2010–2014 **Ph.D. in Computer Science** King's College London, UK

***Thesis**: Persuasion Dialogues & Opponent Modelling.*

Developed logical inference methods for large **Knowledge Graphs**, combining symbolic reasoning, **graph analytics**, and Bayesian techniques (e.g., MCMC). Received **Best Poster Award** at IJCAI 2013 (413 submissions), **EPSRC PhD Scholarship**, **Outstanding TA Award**, and **Graduate Certificate in Academic Practice**.

Access Publications on Google Scholar

2004–2010 **Diploma (BEng) in Computer Engineering** University of Thessaly, GR

Five-year polytechnic degree in Computer Science and Engineering with a **strong focus on mathematics** (10 courses), covering theoretical aspects, networking, and telecommunications. ***Majored in Artificial Intelligence***.

2002–2004 **Military Officer: Second Lieutenant** National Guard, CY

Completed national service as a commissioned Second Lieutenant.

# research & AI standards

since 10/2024 **University College London, Dept. of Information Studies** Associate Researcher

Research/teach practical AI standardisation: AI Act, auditability, model/data versioning, explainability, and safe adoption.

since 2021 **ISO - International Organisation for Standardisation** Committee Expert Member for JTC'21

Member of JTC'21 Working Group 3, developing AI standards aligned with EU policy and international norms.

# selected AI/platform tooling

**Promet** Personal project | Local AI harness

Personal AI assistant runtime exploring controlled tool execution, schema validation, approval boundaries, durable state, traces and replay/evaluation. Hands-on work on the application controls around model behaviour, with explicit permissions and inspectable execution evidence.

PyPI **skeleton-replay** Runtime evidence

Python developer tool that turns script/pytest runs into traces, architecture snapshots, workflow evidence, and replayable reports for review, debugging, onboarding, and evidence-led code understanding.

JetBrains **Skeleton Replay plugin** IDE workflow

Free PyCharm/IntelliJ plugin bringing Skeleton reports, workflow evidence, and source navigation into the IDE; reflects a bias for developer experience, evidence, and shared understanding.

PyPI **dynamicio** Data contracts

Created a Python library for explicit I/O boundaries, schema/data validation and local/dev/prod dataset switching. Used across 15+ repositories to make ML/data workflows easier to test and review.

# talks/interviews

2023 **Bill Raymonds, Agile in Action** Podcast Interview

**Agile Data Science**: Sailing Through Data Science: The Agile Journey at Vortexa, **Details:** YouTube

2022 **Open Data Science Conference** Industry Conference Talk

**dynamicio**, a published **PyPI library** for abstracting I/O in ML/data workflows, **Details:** here

2020 **iunera: Big Data Innovators** Interview Blog Post

The Agile approach in data science explained by an ML expert, **Details:** here

2020 **Big Data Warsaw** Industry Conference Talk

Monitoring & Analysing Communication and Trade Events as Graphs, **Details:** here

2018 **Connected Data London** Panelist @ Industry Conference

AI tribes and interconnections: what's graph got to do with it?, **Details:** here

2018 **Minds Mastering Machines** Industry Conference Talk

A talk on doing Data Science the Agile Way, **Details:** here

# certificates & awards

2017–2020 **Selected ML/cloud credentials** AWS, Google Cloud, O'Reilly

AWS ML Specialty; Google Professional Data Engineer; Apache Spark Developer.

2016–2019 **Data/graph engineering credentials** Coursera, Elastic, Neo4j

Process Mining; Graph Analytics; Advanced Elasticsearch; Neo4j Certified Professional.